

Current-vaccination-rate fits: with vs. without the transmission multiplier $m(t)$

Median [95% interval] across posterior draws (one deterministic simulation per draw). Both models start with no initial immunity and use the current vaccination schedule (`model_config_MA_vax.json`).

- **With $m(t)$** : the fit with the 14-day-knot transmission multiplier (`fitted_params_MA_vax.json`).
- **Without $m(t)$** : $m(t) = 1$ throughout; `beta_baseline`, `humidity_impact`, `seed_scale_E` and `IHR_scale` refit (`fitted_params_MA_vax_no_transmission_multiplier.json`).

Hospitalizations averted: *infection protection only* = no vaccination → infection-protection-only scenario; *hospitalization reduction* = infection-protection-only → full baseline; *total* = no vaccination → full baseline (the sum of the two). Percentages are of the no-vaccination hospitalizations. Attack rate = $(N - S(T) - SV(T)) / N$. Effective IHR = fitted `IHR_scale` × `I_to_H_prop` (unvaccinated) or × `IV_to_H_prop` (vaccinated).

Fitted parameters

Parameter	With $m(t)$	Without $m(t)$
<code>beta_baseline</code>	0.0303 [0.02514 – 0.03736] (best 0.0281)	0.03684 [0.03125 – 0.04106] (best 0.03725)
<code>humidity_impact</code>	0.6033 [0.1422 – 0.9476] (best 0.8809)	0.2061 [0.01192 – 0.5411] (best 0.2175)
<code>seed_scale_E</code>	1.569 [0.37 – 3.721] (best 1.471)	2.034 [0.6519 – 7.337] (best 1.632)
posterior draws / simulations per scenario	627 / 627	429 / 429

Doses delivered (baseline)

Age group	With $m(t)$	Without $m(t)$
0	16,726 [16,639 – 16,803]	15,846 [15,707 – 15,980]
1-4	133,722 [132,798 – 134,147]	125,464 [124,271 – 126,635]
5-12	213,727 [211,360 – 214,867]	191,579 [188,684 – 194,578]
13-17	147,131 [145,737 – 147,810]	133,826 [132,113 – 135,591]
18-49	802,991 [794,183 – 807,168]	724,366 [713,512 – 734,969]
50-64	525,012 [520,409 – 527,185]	484,127 [478,167 – 489,904]
65+	720,282 [717,097 – 721,754]	690,909 [686,327 – 695,212]
All	2,559,590 [2,538,188 – 2,569,715]	2,366,155 [2,339,359 – 2,392,645]

Hospitalizations averted, infection protection only

Age group	With m(t)	Without m(t)
0	40 [24 – 52]	11 [7 – 15]
1-4	208 [140 – 261]	60 [44 – 81]
5-12	188 [126 – 251]	43 [29 – 58]
13-17	115 [74 – 158]	25 [17 – 34]
18-49	743 [548 – 950]	138 [117 – 182]
50-64	792 [584 – 981]	154 [116 – 202]
65+	4,433 [3,698 – 5,162]	697 [537 – 892]
All	6,583 [5,440 – 7,404]	1,132 [914 – 1,383]

% of no-vax hospitalizations averted, infection protection only

Age group	With m(t)	Without m(t)
0	52.7% [48.2% – 55.3%]	20.7% [18.4% – 23.2%]
1-4	56.9% [52.5% – 59.5%]	26.2% [24.0% – 28.7%]
5-12	54.7% [49.4% – 57.7%]	20.7% [18.4% – 23.2%]
13-17	54.7% [49.5% – 57.8%]	20.7% [18.4% – 23.2%]
18-49	49.8% [44.6% – 52.7%]	15.1% [12.8% – 17.7%]
50-64	50.7% [45.9% – 53.5%]	17.0% [14.7% – 19.6%]
65+	48.1% [43.5% – 50.9%]	14.1% [11.7% – 16.8%]
All	49.2% [44.4% – 52.0%]	15.3% [13.0% – 18.0%]

Hospitalizations averted, hospitalization reduction

Age group	With m(t)	Without m(t)
0	0 [0 – 1]	0 [0 – 1]
1-4	4 [2 – 5]	4 [3 – 5]
5-12	3 [2 – 3]	3 [2 – 4]
13-17	2 [1 – 2]	2 [1 – 2]
18-49	11 [9 – 14]	12 [10 – 14]
50-64	16 [13 – 20]	16 [13 – 20]
65+	720 [647 – 801]	627 [528 – 751]

Age group	With m(t)	Without m(t)
All	756 [681 – 837]	663 [562 – 788]

% of no-vax hospitalizations averted, hospitalization reduction

Age group	With m(t)	Without m(t)
0	0.5% [0.5% – 0.5%]	0.8% [0.8% – 0.9%]
1-4	1.0% [0.9% – 1.1%]	1.7% [1.6% – 1.8%]
5-12	0.7% [0.7% – 0.8%]	1.3% [1.2% – 1.4%]
13-17	0.8% [0.7% – 0.9%]	1.4% [1.3% – 1.5%]
18-49	0.8% [0.7% – 0.8%]	1.3% [1.2% – 1.4%]
50-64	1.1% [1.0% – 1.2%]	1.8% [1.7% – 1.9%]
65+	7.7% [7.3% – 8.4%]	12.6% [12.0% – 13.2%]
All	5.6% [5.2% – 6.2%]	9.0% [8.3% – 9.6%]

Hospitalizations averted, total

Age group	With m(t)	Without m(t)
0	40 [24 – 53]	11 [7 – 16]
1-4	212 [142 – 266]	64 [47 – 86]
5-12	191 [128 – 255]	45 [31 – 62]
13-17	116 [75 – 160]	26 [18 – 36]
18-49	754 [558 – 963]	149 [129 – 195]
50-64	809 [599 – 1,000]	170 [130 – 221]
65+	5,147 [4,398 – 5,959]	1,331 [1,102 – 1,596]
All	7,327 [6,169 – 8,190]	1,802 [1,527 – 2,126]

% of no-vax hospitalizations averted, total

Age group	With m(t)	Without m(t)
0	53.2% [48.7% – 55.8%]	21.5% [19.3% – 24.0%]
1-4	57.9% [53.5% – 60.4%]	27.9% [25.8% – 30.4%]
5-12	55.4% [50.2% – 58.4%]	22.0% [19.7% – 24.5%]
13-17	55.5% [50.3% – 58.5%]	22.1% [19.8% – 24.5%]
18-49	50.5% [45.5% – 53.4%]	16.4% [14.1% – 19.0%]

Age group	With m(t)	Without m(t)
50-64	51.8% [47.0% – 54.5%]	18.8% [16.5% – 21.3%]
65+	55.9% [51.8% – 58.2%]	26.8% [24.8% – 29.0%]
All	54.8% [50.5% – 57.4%]	24.2% [22.2% – 26.7%]

Hospitalizations, baseline

Age group	With m(t)	Without m(t)
0	36 [22 – 48]	42 [27 – 58]
1-4	153 [104 – 201]	167 [120 – 217]
5-12	156 [111 – 204]	162 [112 – 208]
13-17	95 [63 – 123]	93 [64 – 125]
18-49	746 [582 – 895]	771 [653 – 933]
50-64	753 [595 – 911]	740 [589 – 916]
65+	4,125 [3,700 – 4,555]	3,630 [3,076 – 4,335]
All	6,070 [5,569 – 6,549]	5,622 [4,943 – 6,389]

Hospitalizations, no vax

Age group	With m(t)	Without m(t)
0	76 [47 – 101]	53 [34 – 73]
1-4	364 [248 – 461]	231 [170 – 303]
5-12	348 [239 – 454]	207 [144 – 270]
13-17	212 [140 – 283]	119 [83 – 160]
18-49	1,496 [1,165 – 1,843]	918 [794 – 1,111]
50-64	1,563 [1,225 – 1,891]	911 [731 – 1,128]
65+	9,255 [8,272 – 10,460]	4,959 [4,184 – 5,879]
All	13,370 [11,947 – 14,577]	7,438 [6,528 – 8,416]

Attack rate, baseline

Age group	With m(t)	Without m(t)
0	4.8% [3.6% – 7.4%]	26.9% [23.7% – 30.0%]
1-4	5.3% [3.9% – 8.1%]	30.0% [26.7% – 33.4%]
5-12	8.1% [6.0% – 12.3%]	44.3% [40.1% – 48.3%]

Age group	With m(t)	Without m(t)
13-17	8.3% [6.2% – 12.6%]	45.1% [40.8% – 49.1%]
18-49	7.6% [5.7% – 11.7%]	41.0% [36.6% – 45.2%]
50-64	6.5% [4.9% – 10.1%]	35.7% [31.7% – 39.7%]
65+	4.1% [3.0% – 6.4%]	24.0% [21.0% – 27.1%]
All	6.7% [5.0% – 10.4%]	36.9% [32.9% – 40.8%]

Attack rate, no vax

Age group	With m(t)	Without m(t)
0	10.1% [7.9% – 14.3%]	33.8% [30.7% – 36.8%]
1-4	12.2% [9.6% – 17.1%]	40.7% [37.3% – 43.9%]
5-12	17.8% [14.1% – 24.4%]	55.7% [52.1% – 59.2%]
13-17	18.3% [14.6% – 25.1%]	56.8% [53.1% – 60.2%]
18-49	15.2% [12.0% – 21.2%]	48.2% [44.4% – 51.9%]
50-64	13.3% [10.4% – 18.6%]	43.0% [39.3% – 46.5%]
65+	7.9% [6.1% – 11.3%]	27.9% [25.2% – 30.7%]
All	13.8% [10.8% – 19.2%]	44.3% [40.8% – 47.7%]

IHR, unvaccinated (I_to_H)

Age group	With m(t)	Without m(t)
0	1.077% [0.594% – 1.366%]	0.226% [0.142% – 0.312%]
1-4	1.086% [0.599% – 1.375%]	0.201% [0.147% – 0.269%]
5-12	0.322% [0.188% – 0.473%]	0.061% [0.042% – 0.081%]
13-17	0.278% [0.151% – 0.444%]	0.051% [0.036% – 0.069%]
18-49	0.331% [0.201% – 0.478%]	0.063% [0.057% – 0.079%]
50-64	0.823% [0.489% – 1.217%]	0.149% [0.116% – 0.191%]
65+	9.712% [6.316% – 12.951%]	1.459% [1.197% – 1.766%]
All	n/a	n/a

IHR, vaccinated (IV_to_H)

Age group	With m(t)	Without m(t)
0	0.982% [0.542% – 1.246%]	0.206% [0.130% – 0.285%]

Age group	With m(t)	Without m(t)
1-4	0.990% [0.547% – 1.254%]	0.183% [0.134% – 0.246%]
5-12	0.294% [0.172% – 0.432%]	0.056% [0.039% – 0.074%]
13-17	0.254% [0.138% – 0.405%]	0.047% [0.032% – 0.063%]
18-49	0.301% [0.183% – 0.435%]	0.058% [0.052% – 0.072%]
50-64	0.750% [0.446% – 1.109%]	0.136% [0.105% – 0.174%]
65+	6.701% [4.358% – 8.936%]	1.007% [0.826% – 1.219%]
All	n/a	n/a